Crypto Performance Dashboard

Analytical Report

BTC, ETH, BNB, SOL, XRP — 365-Day Return & Risk Analysis

PYTHON → SQL → POWER BI

ROLLING 365-DAY PERIOD

5 MAJOR CRYPTOCURRENCIES

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Project Overview

Central Business Question

Which cryptocurrency delivers the **best return-to-risk ratio** over the last 365 days?

Business Context

This project analyzes the performance of five major cryptocurrencies — BTC, ETH, BNB, SOL, and XRP — over a rolling 365-day period. Data was collected via API calls and processed through a complete analytical pipeline.

Audience

Managers, Directors — non-technical profiles

Analytical Pipeline

01

Data Collection

API Call — raw crypto prices & metrics

02

Data Preparation

Python (Pandas) — cleaning & engineering

03

Data Analysis

SQL — MySQL Workbench — 10 queries

04

Data Visualization

Power BI — 2-page decision dashboard

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Dataset Summary

Two datasets collected via API, covering 365-day aggregated metrics and daily historical prices across all five coins.

Dataset	Rows	Columns	Period
crypto_metrics	5	7	365-day aggregated
historical_crypto_prices	395	24	April 2025 → April 2026

Variable Categories

Measure

annual_return_pct, risk_level_pct, return_risk_ratio, price_range_usd, daily returns

Segmentation

risk_category, performance_label, return_direction, crypto_name

Temporal

Date, month, quarter, year

Data Quality Treatments

- **1 row dropped:** missing return values (structural — no prior date)
- **Outliers on prices** (BTC ~$124k): kept — real market events
- **Date column:** converted from text to datetime
- **Prices normalized** Min-Max (0–1) for cross-coin comparison

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Python EDA Pipeline

Step 1 — Data Collection & Loading

Both datasets loaded from API into pandas DataFrames. Initial exploration confirmed 5 coins, 396 daily observations, and 11 raw variables.

Step 2 — Data Cleaning

Date converted to datetime format. First row removed (no calculable return without a prior date). Prices normalized Min-Max to enable cross-coin visual comparison.

Step 3 — Feature Engineering (11 new variables)

risk_category (Low / Medium / High Risk), performance_label (Efficient / Inefficient), price_range_usd, month/quarter/year segmentation, and return_direction per coin per day.

Step 4 — Export to MySQL

Final datasets exported as CSV and imported into MySQL via SQLAlchemy for SQL analysis.

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Key SQL Findings

QUERIES Q1 – Q5

Q1 — Best Annual Return

ETH: +36.73% — only coin with a positive annual return. BNB follows at +13.03%. BTC, SOL, and XRP are negative.

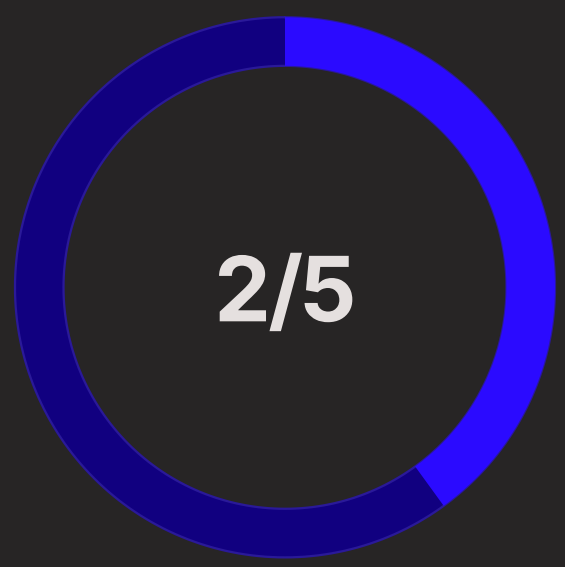
Q2 — Lowest Risk

BTC: 42.47% volatility → Low Risk. BNB: Medium Risk (50.22%). ETH, XRP, SOL: High Risk (>65%).

Q3 — Best Return/Risk Ratio

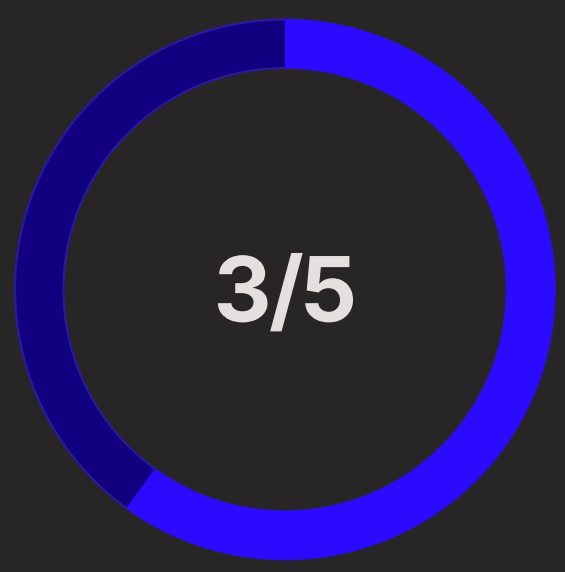
ETH leads with a Sharpe ratio of 0.506. Only ETH and BNB have positive (Efficient) ratios.

Q4 — Efficient vs Inefficient Coins



Efficient Coins

Avg return: **+24.88%**



Inefficient Coins

Avg return: **-14.02%**

Q5 — Price Amplitude (USD Range)

BTC

$61,919 range — extreme exposure

ETH

$3,357 range

BNB

$758 range

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More SQL Findings

QUERIES Q6 – Q10

Q6 — Best Avg Daily Return

ETH: +0.00100/day (~+0.10%) — best daily performer. BTC and SOL are slightly negative.

Q7 — Best BTC Quarter

2025Q2 is the best quarter for BTC: avg return +0.00294/day. 2025Q4 and 2026Q1 are negative — bearish phase.

Q8 — Most Bullish Days

BNB: 199 bullish days (most consistent upward momentum). ETH: 189 | SOL: 182 | BTC: 181 | XRP: 174.

Q9 — Risk vs Return Relationship

BTC = lowest risk but negative return → risk is NOT automatically rewarded. ETH = high risk but the only strongly positive return. This challenges conventional risk-reward assumptions.

Q10 — Daily Volatility (Std Dev)

Coin	Std Dev	Profile
SOL	0.0382	Most volatile
ETH	0.0380	High volatility
XRP	0.0362	High volatility
BNB	0.0263	Medium
BTC	0.0222	Most stable

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Dashboard Page 1: Performance Overview

Crypto Performance Overview: BTC, ETH, BNB, SOL, XRP (365 Days)

36.73%

Best Annual Return

ETH — only positive annual return

0.51

Best Efficiency Score

ETH Sharpe Ratio

2/5

Efficient Coins

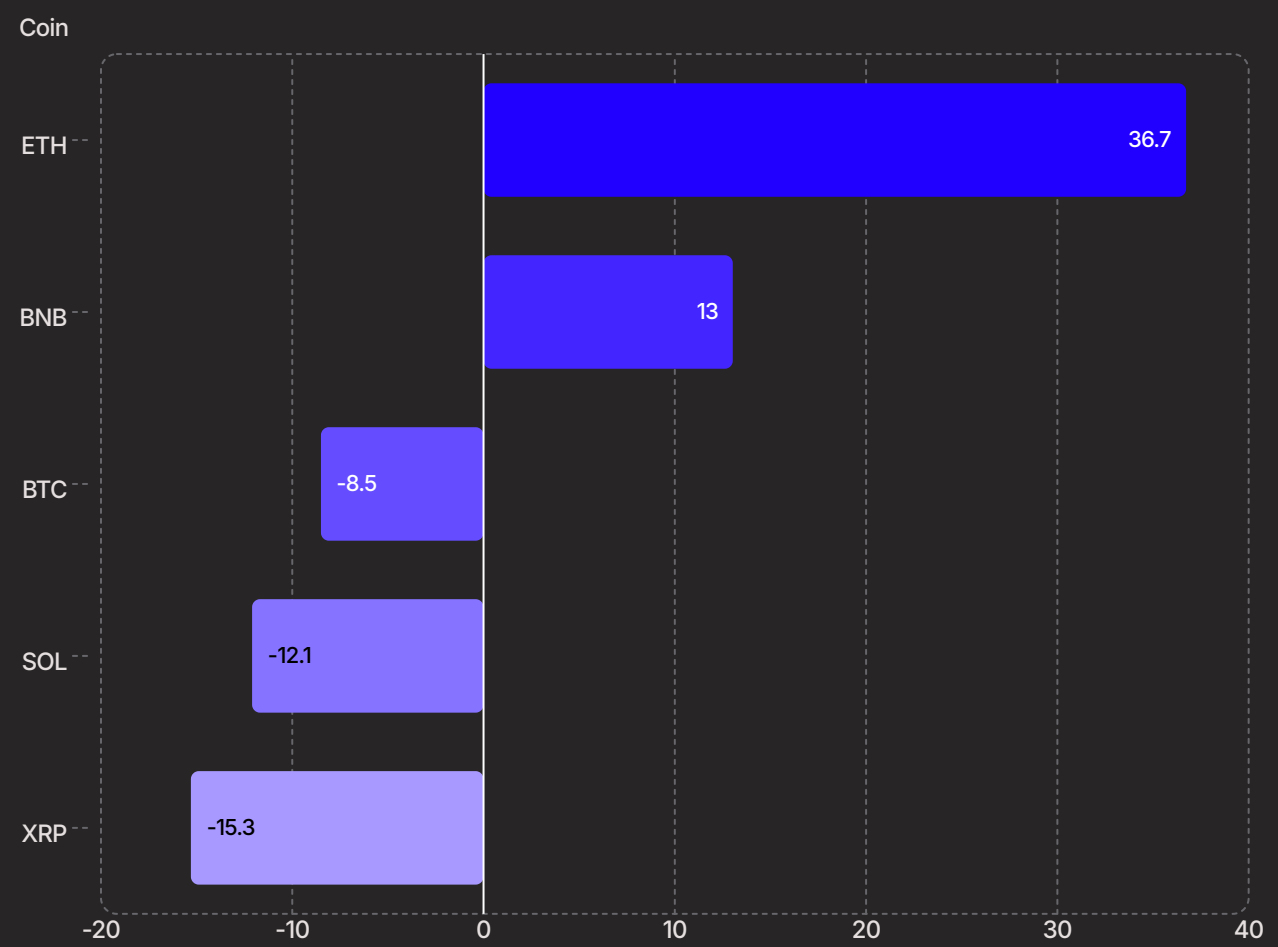
ETH and BNB only

61.53%

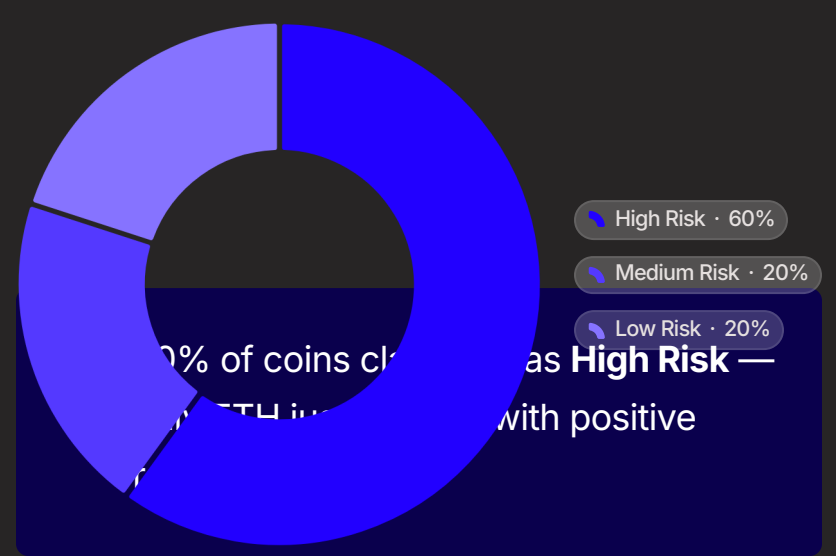
Avg Risk Level

Across all 5 coins

Annual Return by Coin



Risk Category Distribution



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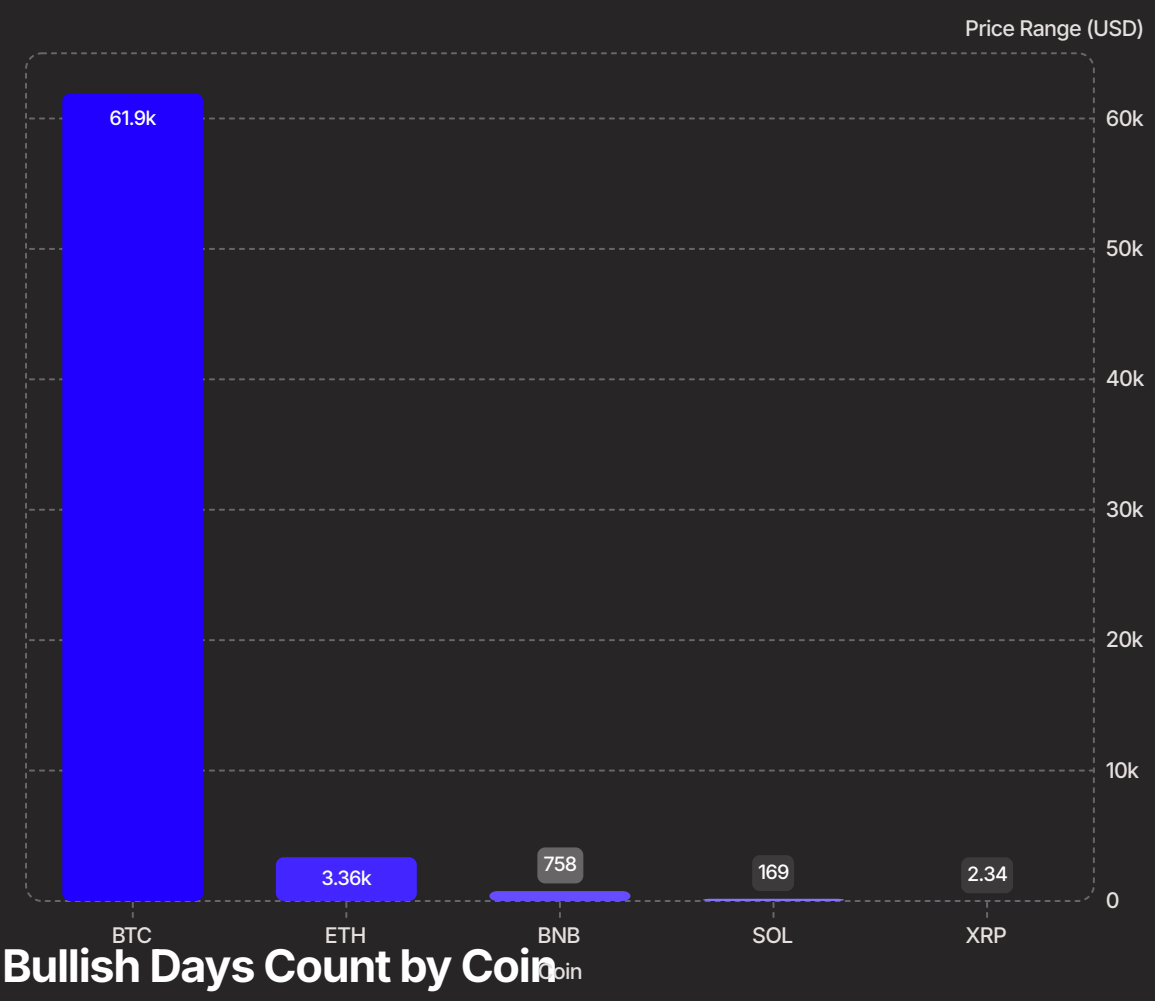
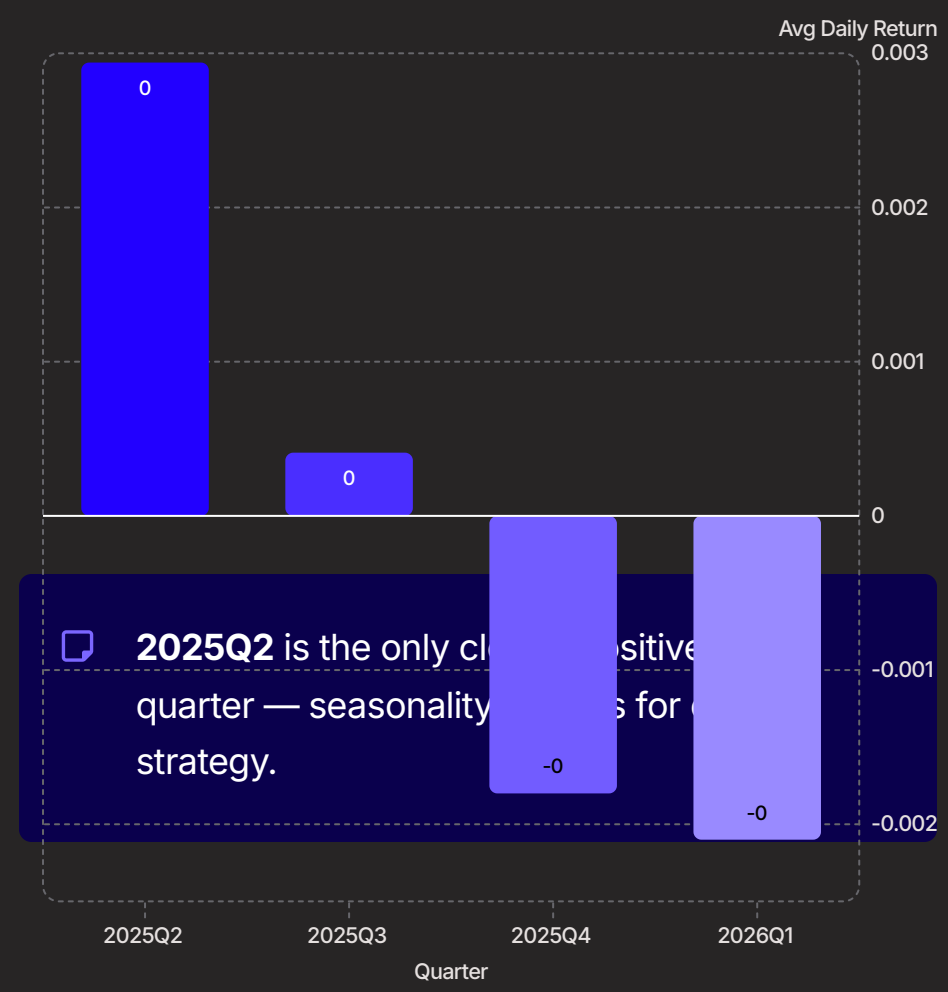
Dashboard Page 2: Price & Trend Analysis

Price And Trend Analysis: BTC, ETH, BNB, SOL, XRP (365 Days)

Price Range Amplitude by Coin (USD)

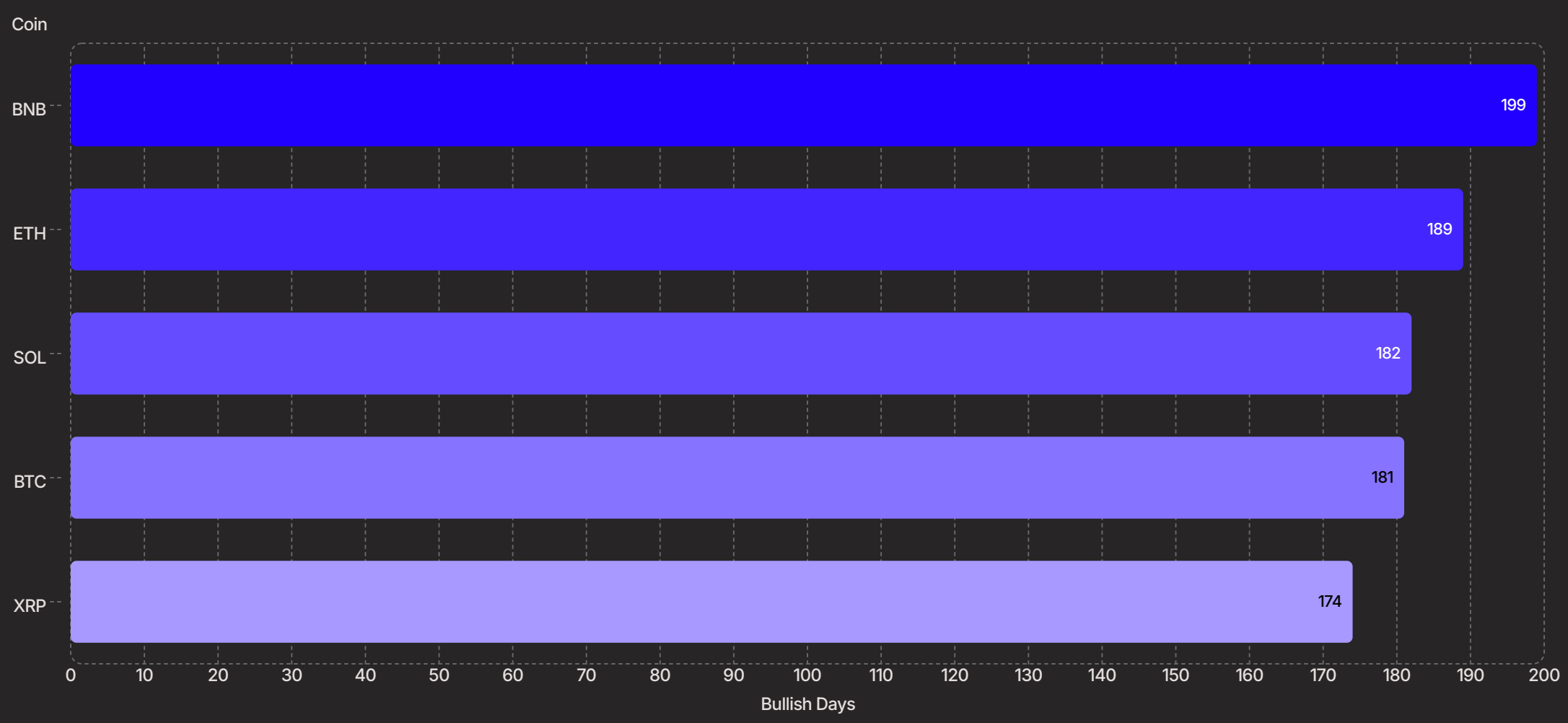


BTC Average Daily Return by Quarter



2025Q2 is the only cl ositive quarter — seasonality s for strategy.

Bullish Days Count by Coin



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Business Recommendations

🟢 Prioritize ETH

Only coin with positive return AND positive Sharpe (0.506). **Allocate dominant weight** to ETH in any active strategy.

🟡 Add BNB as Secondary

Most bullish days (199) + Efficient classification. **Use BNB** to reinforce portfolio consistency.

🔵 BTC = Defensive Only

Lowest risk but negative return. **Reserve BTC** for safe-haven profiles, not performance drivers.

Avoid SOL & XRP (Short-Term)

3 out of 5 coins are Inefficient. **Monitor SOL and XRP** for next quarter before re-entry into active allocation.

Integrate Seasonality

2025Q2 = only clearly positive BTC quarter. Prioritize BTC positions at the start of Q2 if the seasonal pattern confirms in future cycles.

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Analytical Limits, Assumptions & Pipeline Summary

Analytical Limits & Assumptions

Rolling Window

Conclusions apply to this specific 365-day period — not generalizable to other periods.

Simplified Sharpe

Risk-free rate assumed at 0 in Sharpe calculation.

No Inter-Coin Correlation

Dashboard does not analyze diversification benefit of a multi-coin portfolio.

Market Data Only

No macro factors (regulation, adoption, news) are integrated.

Past Performance

No predictive model — analysis is purely descriptive and historical.

Full Analytical Pipeline

1

API Call

Raw data collection — crypto prices & metrics

2

Python (Pandas)

Data cleaning, feature engineering, normalization, export

3

SQL — MySQL Workbench

10 analytical queries: return, risk, efficiency, seasonality, behavioral patterns

4

Power BI

2-page decision dashboard with KPI cards, scatter plot, stacked bar, line chart, slicers

DATA WRANGLING | FEATURE ENGINEERING | SQL ANALYTICS | DAX MEASURES | DASHBOARD DESIGN | BUSINESS STORYTELLING

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